

1 Introduction

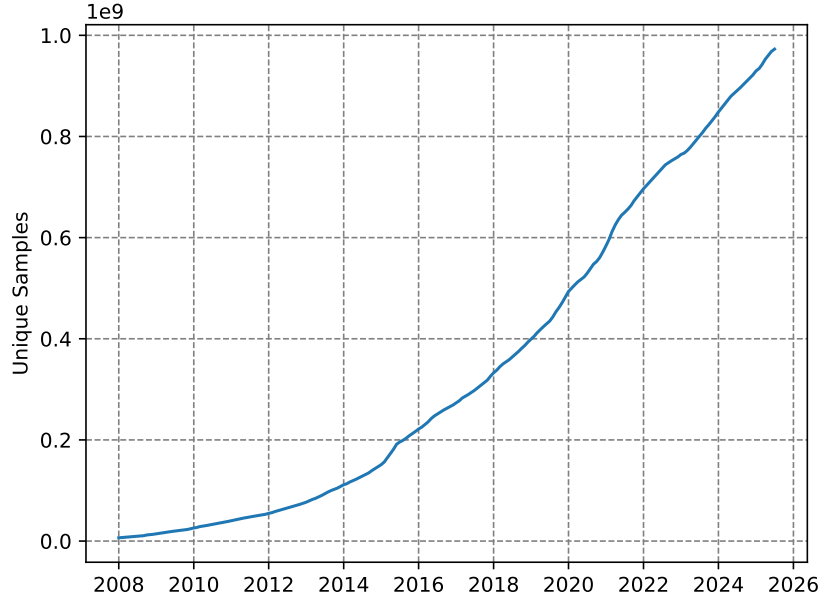


Figure 1: Observed growth in the number of unique forms of malware over time. Data taken from AV-TEST - The Independent IT-Security Institute, 2025

Malicious software, also known as malware, may be defined as software that has an adverse impact on the expected operation of a computer system. This encompasses the provision of access by an unauthorised user to an infected host, the illegitimate extraction and modification of information, or the denial of access to its users. They are considered one of the most pervasive threats to both individuals and organisations in our current digital threat landscape (European Union Agency for Cybersecurity, 2024; Alenezi et al., 2020).

The threat posed by malware can be seen in cyberattacks such as the 2022 Viasat hack (Quiquet, 2023), and the 2017 WannaCry incident (National Audit Office (NAO), 2017; Akbanov et al., 2019). In both of these cases, sophisticated malware were used to disrupt critical infrastructure including communications infrastructure across Europe, and the infrastructure underpinning the National Health Service of the United Kingdom.

In recent years, both the sophistication (Alenezi et al., 2020; Baezner et al., 2017) and frequency (Figure 1) of malware have increased significantly. As a result of both this, and the potential of malicious software to cause significant damage to digital systems, there has been a great deal of interest into how we can reliably identify malware (M. et al., 2023; Maniriho et al., 2024; Aboaoja et al., 2022). In this literature review, we explore common features of malicious software, how sophisticated modern malware avoid detection by conventional means, and both historical and contemporary attempts at identifying such programs.

2 Malware

As outlined in the previous section, malware are a form of software that have a detrimental impact on the expected operation on any system it is installed. Both ours, and more formal definitions, such as the one presented by (Kramer et al., 2010), which define malware as software that cause other software to deviate from their intended behavior, typically in a harmful manner, fail to define what is meant by harmful however. Thus, we define harmful behavior as actions that are in violation of the Confidentiality-Integrity-Availability (CIA) triad (Cochran, 2024). This is a key concept of modern information security policies, that outlines the need for a system to be protected against unauthorised access (confidentiality), modification (integrity), and reliable (availability) to be considered secure.

Malware is often categorised according to its behavior, and the means by which it spreads. We present the following taxonomy based on the work of Maniriho et al., 2024 shown in Figure 2 that outlines the different kinds

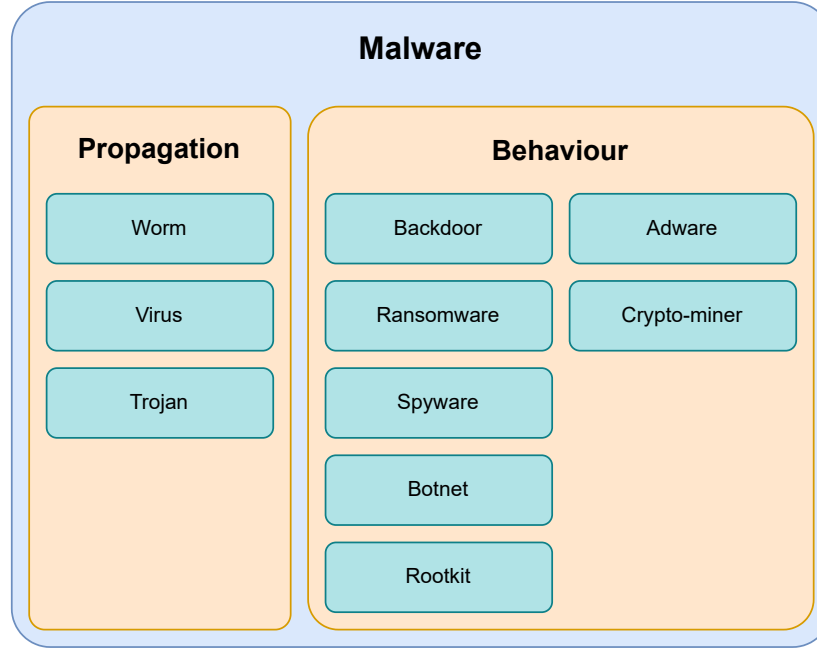


Figure 2: Categories of Malware, including both the means of propagation and behaviour on an infected host.

of malware seen by security researchers, and how they are spread. Note that categories within this taxonomy are not mutually exclusive, as modern malware often exist within multiple families at the same time.

- **Worm** — Malicious software that automatically propagates throughout a network.
- **Virus** — Programs that modify other programs to replicate themselves throughout an infected host.
- **Trojans** — Malware that disguises itself as a legitimate program to infect a host.
- **Backdoor** — Malware that provides covert remote access to infected systems.
- **Ransomware** — Malware that denies access to systems or information until a fee is paid.
- **Spyware** — Malware that spy on the user without their knowledge, often extracting information such as credentials, keyboard logs, and live screenshots of the users activity.
- **Botnet** — Malware that integrate infected systems into a coordinated network of agents.
- **Rootkit** — Malware providing privileged access to an infected system.
- **Adware** — Malware that display advertisements on the host machine.
- **Crypto-miner** — Malware that hijack the users hardware to mine cryptocurrencies.

2.1 Evasion Techniques

In addition to performing these harmful actions, malware also employ various techniques designed to make it difficult to detect by automated security services, and the user. This allows the software to go undetected for as long as possible, maximising the potential amount of damage it can do to the system before it is eventually found. These are known as evasion techniques, which encompasses obfuscation, environmental awareness, and living-off-the-land. In the following section, we discuss these techniques, and why they are effective at defeating existing anti-malware solutions.

2.1.1 Obfuscation

Obfuscation is defined as the act of making something more challenging to understand. In the context of malware, and its detection, it refers to a set of evasion techniques that modify the structure of a program, while maintaining its behavioral characteristics. These techniques are particularly effective at defeating static signature based methods, as they rely upon simple, byte-level patterns associated with malicious programs.

The first of these obfuscation techniques is known as register re-assignment. Registers are small portions of memory within the CPU that store intermediate values during computation. A number of these registers exist, some with a specific purpose and others for more general usage. In the 64-bit x86 instruction set, 16 general purpose registers are present. By changing which registers values are stored in, we modify the structure of a program, without affecting its behavior. Early signature based anti-malware solutions relied on a database containing the cryptographic hashes of malicious programs. Minor changes to the representation of a program, that have no bearing on its behavioral characteristics, such as changing the registers used to store intermediate values, will result in a unique hash. Thus, the program will not be found in the database, and allowed to run, resulting in the infection of a host.

While this approach is successful at defeating early anti-malware solutions, more targeted signature based methods that recognise a sequence of bytes within the program are able to defeat it through the use of an intermediate representation. A technique that is able to defeat these sophisticated methods is to insert instructions that have no impact on the execution of the program. For example, the `nop` instruction tells the CPU to do nothing, and is often used for padding, or timing purposes. However, by inserting these useless instructions, signature based methods that search for the presence of a specific pattern of bytes associated with a certain piece of malware would be unable to recognise the program as malicious. This is known as dead code insertion.

Within an executable program, code are organised into a set of basic blocks, which each have a label associated with them. When the program reaches a conditional operation, the pointer to the next instruction jumps to the location of the selected block, and continues execution. By changing where these basic blocks are located within the program, it is possible to create unique programs that are behaviourally identical to each other. This is a form of obfuscation known as code transposition.

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